NVDLA on PetaLinux 2024.1

Correctness, latency, throughput, and monitored energy at a glance

100 / 100

LeNet stability

One boot; no module reload or PL reset

246 / 246

ResNet-50 hardware layers

Exact match to source-built nv_small VP golden

1.34x

CPU INT8 execution advantage

378.7 ms CPU INT8 vs 507.7 ms NVDLA on ResNet-50

1.92x

NVDLA cold-deployment advantage

1.336 s NVDLA vs 2.566 s CPU INT8 on ResNet-50

-62.3%

NVDLA incremental energy

0.134 J NVDLA vs 0.356 J CPU INT8 on ResNet-50

60

Selected benchmark sessions

Twelve balanced cohorts; five fresh boots each

Comparison boundary: nv_small NVDLA INT8 is compared with both FP32 and independently quantized QDQ INT8 ONNX Runtime on four Cortex-A53 cores. Equal nominal precision does not imply identical quantization or graph transformation. Power is monitored PS + PL rails, not external 12 V input.

Correctness before performance

Layered acceptance

- OK — Pinned inputs
- OK — ABI and build
- OK — Verified nv_small VP
- OK — Driver probe
- OK — GEM mapping
- OK — IRQ + engine completion
- OK — Exact tensor + repeat

LeNet / MNIST

Fast repeat oracle

INPUT **1 x 1 x 28 x 28**	LOADABLE **0.446 MB**	HARDWARE LAYERS **10**
ENGINE MIX **4 Conv / 4 SDP / 2 PDP**	VP **Exact output**	BOARD **Exact; 100/100 repeats**

Output: 0 2 0 0 0 0 0 124 0 0

ResNet-50

Large multi-engine graph

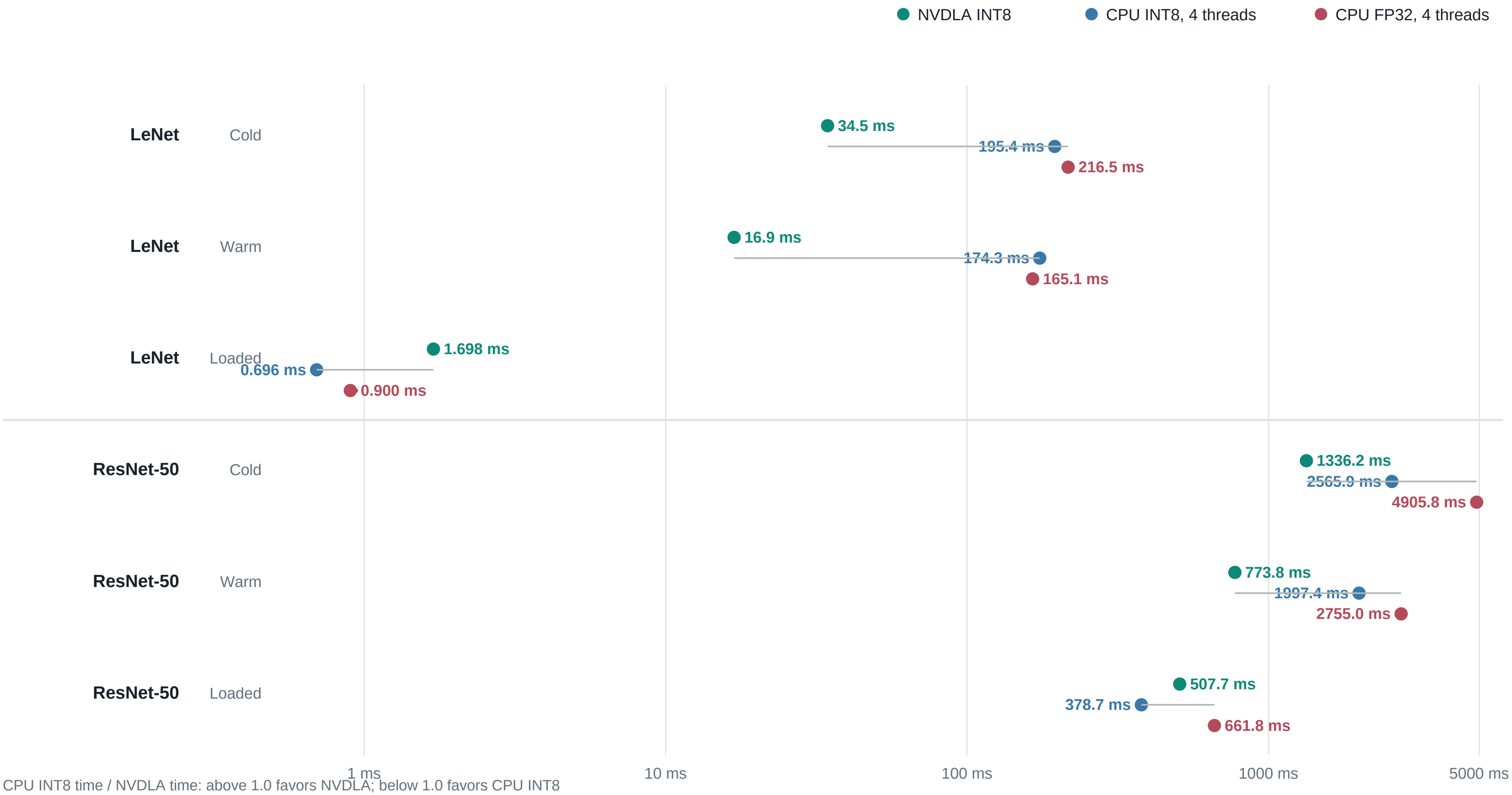
INPUT **1 x 3 x 224 x 224**	LOADABLE **25.77 MB**	HARDWARE LAYERS **246**
ENGINE MIX **114 Conv / 130 SDP / 2 PDP**	VP **246/246; golden established**	BOARD **246/246; exact hash**

Output: 1,000 signed values; SHA-256 842d34f...

Sources: VP configuration audit, differential trace, board stability and ResNet-50 artifacts

Correctness-qualified samples; no outliers discarded

Three deployed stacks across each timing boundary



CPU INT8 time / NVDLA time: above 1.0 favors NVDLA; below 1.0 favors CPU INT8

5.67x	10.33x	0.41x	1.92x	2.58x	0.75x
LeNet cold	LeNet warm	LeNet loaded	ResNet cold	ResNet warm	ResNet loaded

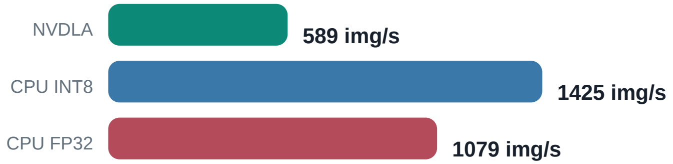
Unpowered cohorts; points are medians across five independent boot-session medians

Correctness-qualified samples; no outliers discarded

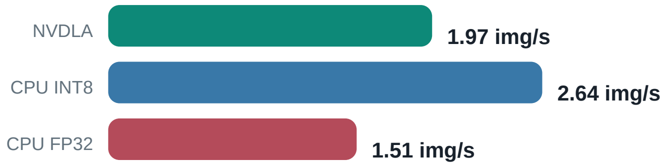
What changes between LeNet and ResNet-50?

Loaded-context throughput

LeNet



ResNet-50



Workload scale and operation mix

LeNet

NVDLA: 0.446 MB loadable | 10 hardware layers

CPU ONNX: 1.73 MB FP32 / 0.45 MB INT8

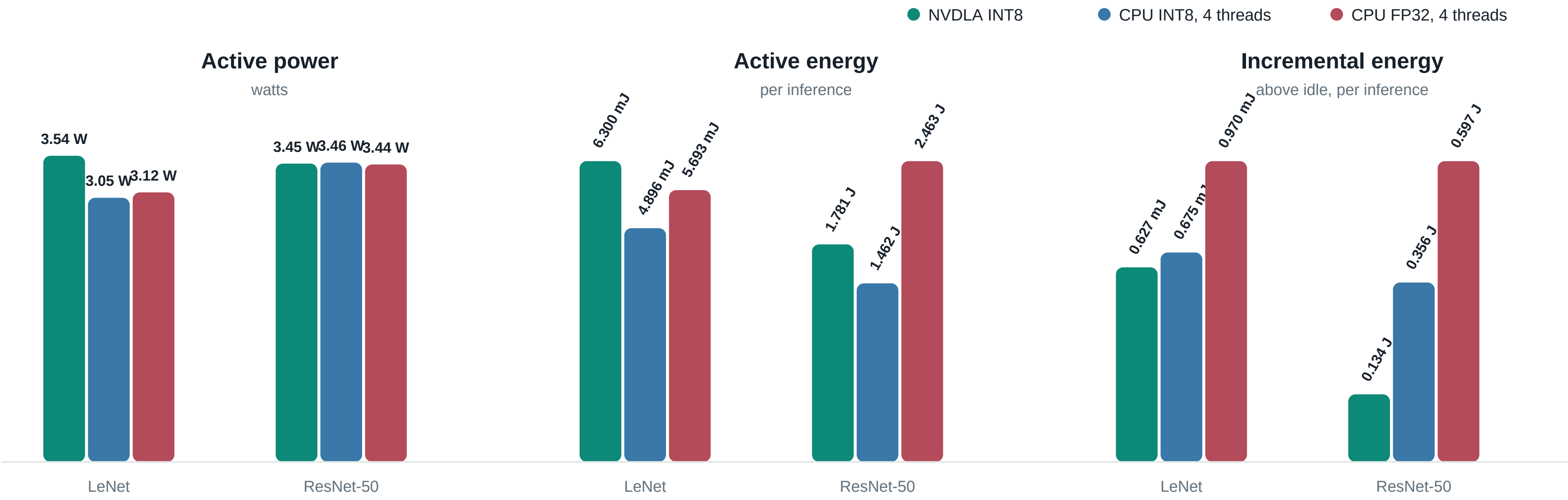
ResNet-50

NVDLA: 25.77 MB loadable | 246 hardware layers

CPU ONNX: 102.48 MB FP32 / 26.09 MB INT8

ResNet-50 has **24.6x** more hardware layers and a **57.8x** larger loadable.
Its larger execution graph amortizes fixed submission and interrupt costs.

Monitored PS + PL rails during inference



LeNet

+28.7% NVDLA active energy

-7.1% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

ResNet-50

+21.8% NVDLA active energy

-62.3% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

What was collected, and how to read it

1. **Fresh boot** — Unique Linux boot ID
2. **Settle + verify** — Clock, frequency, hashes
3. **Correctness gate** — Golden output + kernel health
4. **Measure** — Unpowered latency or powered batch
5. **Archive** — Raw profiles, rails, logs
6. **Aggregate** — Five session medians + 95% CI

Metric inventory

Correctness	Output hashes, exact tensors, engine sequence, IRQ deltas, repeat failures
Latency	Cold deployment, warm deployment, loaded-context execution, runtime phases
Power	18 PS + PL rails, idle/active/incremental watts, integrated energy/inference
Statistics	Raw samples, mean/median, spread, percentiles, retained outliers, bootstrap CI
Workload	Input shape, loadable/model size, HWL count, per-engine operation count
Throughput	Images/s for every timing boundary; CPU/NVDLA latency ratios
Environment	Kernel, binary hashes, clock, CPU affinity/frequency/governor, boot ID
Evidence	Serial, dmesg, runtime profiles, sensor samples, manifests, selection hashes

Interpret with: one ZCU102 and nv_small implementation; NVDLA INT8 versus independently quantized CPU INT8 plus an FP32 reference; monitored rails rather than wall input; five independent boots per cohort.